

Exercise Set 2

17 September 2025

Syntax $\neq$ Semantics

1 Syntax

Given the following program

```
remainder(a,b) =  
  aux := b;  
  while (res >= aux)  
    res := res - aux;  
  done  
  res;
```

- Which syntactic error(s) can you spot immediately? Your answer should make distinct errors due to the context free grammar from those due to (implicit) contextual verification rules.

2 Semantics

Given the following programs

<pre>abs(a,b) = res := 0; if (a < b) then res := b - a else res := a - b endif; res</pre>	<pre>fib(n) = if (n = 0) then res := 0 else if (n = 1) then res := 1 else res := fib(n-1) + fib(n-2) endif endif; res</pre>	<pre>gcd(a,b) = while (a != b) do if (a > b) then a := a - b endif done; a</pre>
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- Express for each function what they *should* compute.
- Which preconditions need to be ensured on the different programs in order to be semantically correct with respect to your previous answer?
- Given the following memory state $\mu = \{a \leftarrow 3, b \leftarrow 9\}$, evaluate the operational semantics of the programs **abs**(a, b), **fib**(a) and **gcd**(a, b)

3 Modelling

In the following, we introduce arrays into our “language”. By convention, given an array $\mathbf{a}$, we will use the following syntax elements

- $\mathbf{a}[\mathbf{i}]$: access to the element in position $\mathbf{i}$ of the array $\mathbf{a}$
- $\mathbf{i} \in \{0, \dots, n - 1\}$: the indexing of the array starts by default at 0 and ends at $n - 1$ (if n is the size of the array)
- $\mathbf{a.length}$: when the dimension of the array is not explicitly given, you can get the length of the array with the following signature

Here is an informal specification for a function: “Given two arrays a and b of same size, every element of b will be (after executing the function) equal to the sum of the preceding elements (last comprised) of a ”

1. Implement the function accordingly to the previous specification
2. Express the precondition and poscondition of the function using FOL formulas
3. Given $\mu = \{a \leftarrow [0, 3, 5, 6], b \leftarrow [0, 0, 0, 0]\}$, evaluate the operational semantics of the function you wrote